

Lab 9 : Competitive exclusion and predator-prey dynamics

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CS302, Modeling and Simulation
 (Dated: June 23, 2025)

In this lab, we studied the Principle of Competitive Exclusion and modelled the population growth of two similar species. We also modelled predator-prey dynamics with and without human interference.

I. COMPETITIVE EXCLUSION

In Nature it happens often that two similar species compete for resources and living space within the same ecological territory. The outcome of this competition is usually the extinction of one species. This is known as the **Principle of Competitive Exclusion**. Let the population densities of an X-species be $x(t)$ and of a Y-species be $y(t)$.

A. Model

$$\dot{x} = Ax - Bx^2 - \alpha xy \quad (1)$$

$$\dot{y} = Cy - Dy^2 - \beta xy \quad (2)$$

B. Results

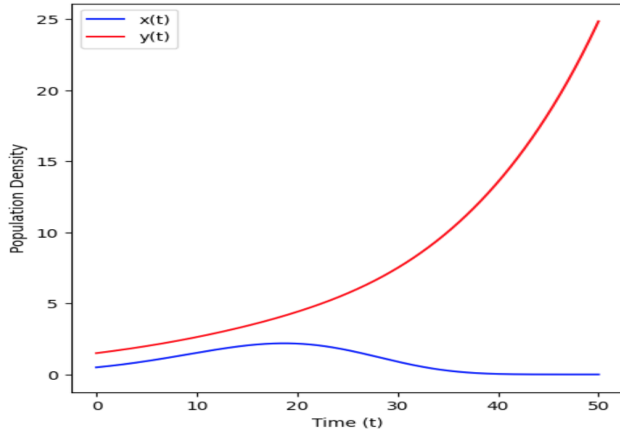


FIG. 1. Population densities of X-species and Y-species vs Time. $A = 0.21827, C = 0.06069, B = 0, D = 0, \alpha = 0.05289, \beta = 0.00459, x_0 = 0.5, y_0 = 1.5, \Delta t = 0.0001$.

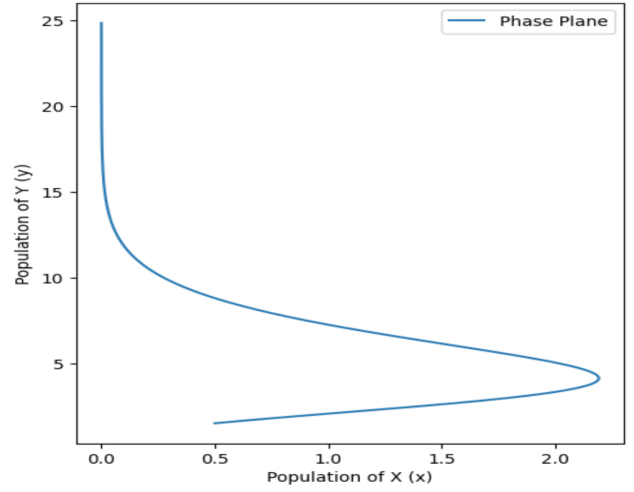


FIG. 2. Phase plot: Y-species vs X-species

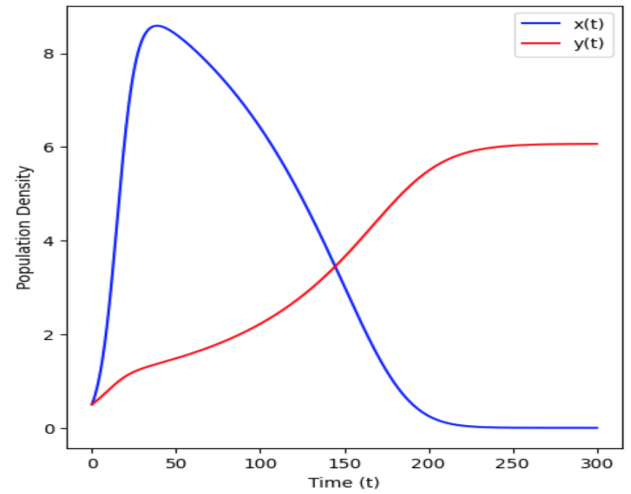


FIG. 3. Population densities of X-species and Y-species vs Time. $A = 0.21827, C = 0.06069, B = 0.017, D = 0.010, \alpha = 0.05289, \beta = 0.00459, x_0 = 0.5, y_0 = 0.5, \Delta t = 0.0001$. peak values X spices 8.58 Y spices 6.06.

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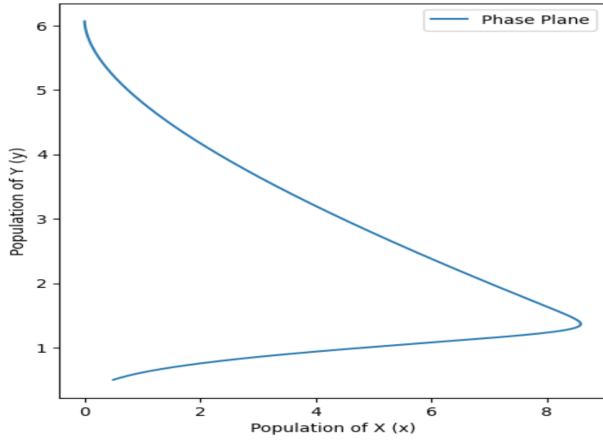


FIG. 4. Phase plot: Y-species vs X-species

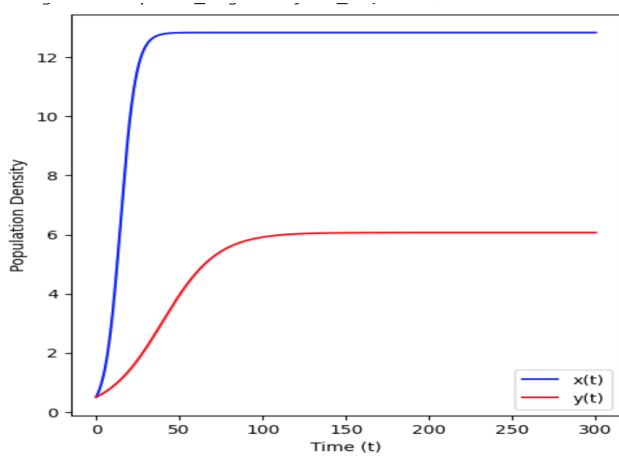
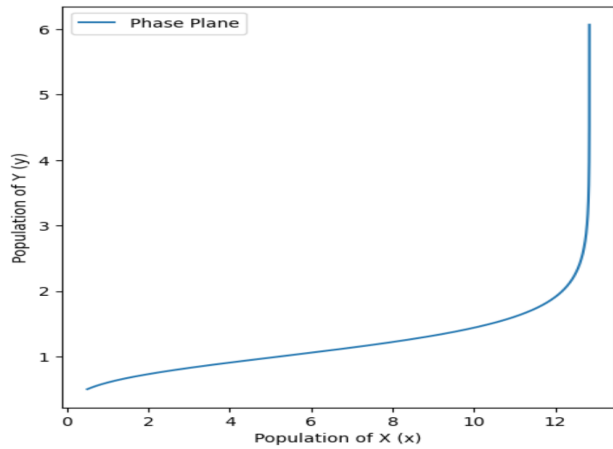
FIG. 5. Population densities of X-species and Y-species vs Time for no competition. $A = 0.21827, C = 0.06069, B = 0.017, D = 0.010, \alpha = 0, \beta = 0, x_0 = 0.5, y_0 = 0.5, \Delta t = 0.0001$. peak values X spics 12.84 Y spics 6.06.

FIG. 6. Phase plot: Y-species vs X-species

II. PREDATOR-PREY DYNAMICS

In the interaction between a prey species X and a predator species Y , let the population densities (number per unit area) of the X -species (prey) be $x(t)$ and of the Y -species (predator) be $y(t)$. The growth of x and y is modelled by the coupled equations

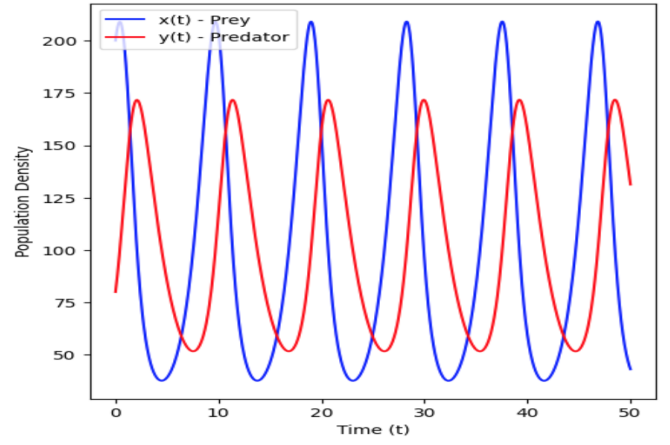
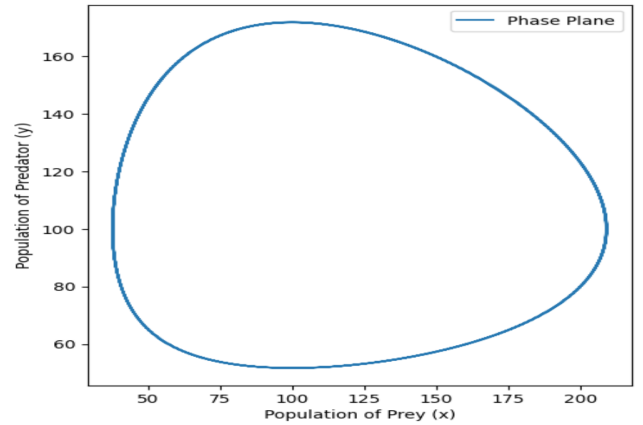
A. Model

$$\dot{x} = Ax - Bxy - \epsilon x \quad (3)$$

$$\dot{y} = -Cy + Dxy - \epsilon y \quad (4)$$

B. Result

Parameters: $A = 1, B = 0.01, C = 0.5, D = 0.005$.

FIG. 7. Population densities of prey X and predator Y vs Time without human interference. $x_0 = 200, y_0 = 80, \epsilon = 0, \Delta t = 0.0001$. peak values pray 209.18 predator 171.71FIG. 8. Phase plot: Predator Y vs Prey X

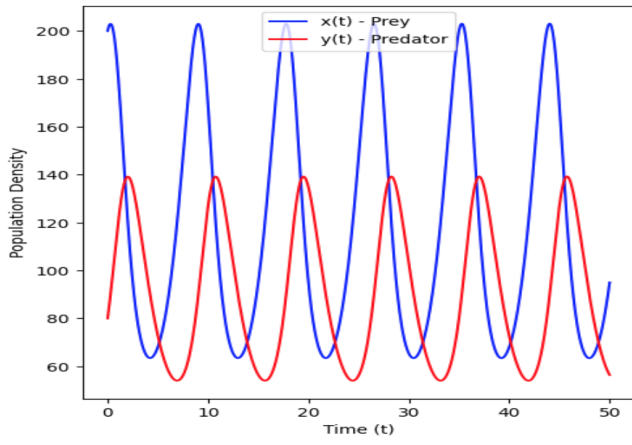


FIG. 9. Population densities of prey X and predator Y vs Time with human interference. $x_0 = 200, y_0 = 80, \epsilon = 0.1, \Delta t = 0.0001$. peak values pray 202.16 predator 139.21

Due to human interference, the peak value of prey decreases by about 8 units, whereas the peak value of predators decreases by about 30 units, indicating that human interference benefits prey.

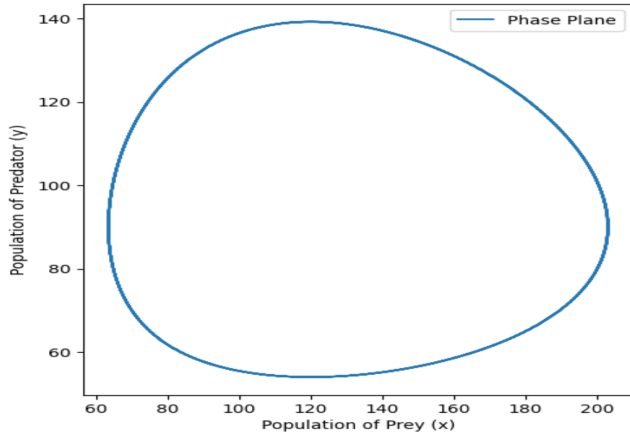


FIG. 10. Phase plot: Predator Y vs Prey X

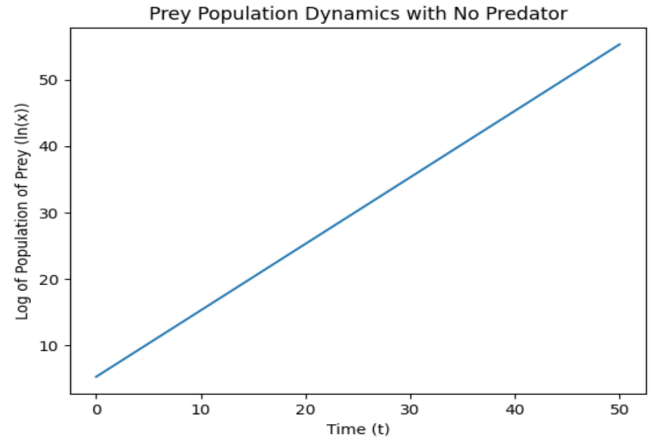


FIG. 11. Population density of prey without predator in logarithmic scale. $x_0 = 200, y_0 = 0, \epsilon = 0$.
without predator prey will increase in time.

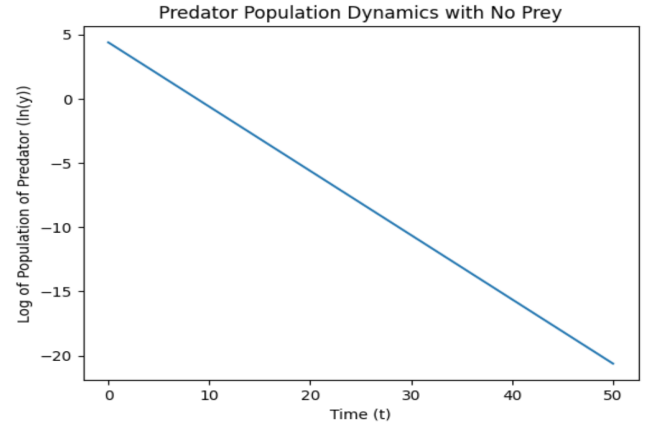


FIG. 12. Population density of predator without prey in logarithmic scale. $x_0 = 0, y_0 = 80, \epsilon = 0$.
without pray predator will eventually extinct.